

Stray Notes: Yann van der Cruyssen on composing soundscapes beyond human trespass

The multidisciplinary creator known as Morusque details some of the thinking, tricks, and trade-offs behind the audio for feline adventure *Stray*. By Jerry Jeriaska of *The Ongaku*

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Stray takes place in a future era, within a dilapidated urban setting where humans are no longer present. Humanoid robots have inherited the cramped apartments and corner stores built by their progenitors, and now instinctually pay homage to a departed civilization by carrying on human customs.

Into this makeshift robot society wanders an unnamed feline who serves as the protagonist of *Stray*. Aided by a friendly aerial drone named B-12, the cat can communicate with the local inhabitants. One robot B-12 encounters insists on burning incense to waft a pleasant smell through their living quarters, admitting wistfully that they lack the sensors to detect the odor. Early on the cat runs into a street musician carrying an electronic facsimile of a simple acoustic guitar fashioned out of an oil canister.

The robot's name "Morusque" will sound familiar to those who have played *Blocks That Matter* (2011), *Tetrobot and Co.* (2013) and *Seasons After Fall* (2016) — a trilogy of puzzle games and platformers by independent developer Swing Swing Submarine, based in Montpellier, France. The mechanical street performer serves as something of an in-game avatar for *Stray*'s music composer Yann van der Cruyssen, who wrote music under the artist name Morusque for Swing Swing Submarine.

Collecting sheet music in the slums



Van der Cruyssen fashioned two distinctive styles of audio tracks for *Stray*, illustrating the robot cast's yearning to salvage the remains of human culture. He mentions one scene that was indicative of these androids' limitations: "Here sometimes an in-game radio would switch from a jazzy tune to a weird noisy loop, as if they were perceived as similar, and the nearby robots would listen anyway without reacting."

These noisy bits the composer constructed partially using a heavily algorithmic process, while songs like Morusque's "Petite Valse," "Ballad of the Lonely Robot," and "The Way You Compute Tonight" were constructed more conventionally on synths. "I tried to imitate music that humans used to do," he says of his in-game counterpart's compositions, "but I wanted it to be somehow distant... Everything is supposed to be a bit rusty."

"That was one of the first things that sold me on this project," the composer recalls. "When Swann (Martin-Raget), the producer, first told me about it he said it would be a world where robots take actions but it doesn't make sense anymore to them. They are just imitating humans."

Morusque is searching the Slums for some misplaced sheet music. Observing the feline's seemingly limitless curiosity, he asks the player character for help in scouring the nearby debris for his lost notes. Retrieving the scattered papers, the feline can lie down on a carpet and nap next to the street musician as he strums out a tune on his peculiar musical instrument.

"I know that I rewrote some of the sentences that he says," recalls Van der Cruyssen, aka Morusque. "Now that I see people playing, I see that people are bringing him various items because they think it has something to do with him. Most of the time he is answering, 'I don't know what this is.' I kind of regret not writing text for every situation, since it turns out to be my character."

The appearance of Morusque in the story of *Stray* was one late-stage addition, appearing years after Van der Cruyssen began work on prototyping with BlueTwelve in 2017. The composer was responsible for establishing the overall sound for the game, taking on music composition, sound design, integration, and even the troublesome task of debugging. During the final six-month firestorm of development, Raphaël Monnin was hired to contribute to sound design on effects and cinematic cutscenes. It was around this same time that a generic robot stand-in was repurposed to become Morusque.

"I think it was Viv's idea," the composer says, referring to BlueTwelve co-founder Vivien Mermet-Guyenet. "Every developer has their own character in the game. When the trailer was released last year, it seemed obvious that I was going to be that guy."

Guided by the player, the cat protagonist can sidle up to the leg of a robot and snuggle against the machine's metallic limb, eliciting a "smile" on the face of the motorized stranger. The robots of *Stray* have cathode-ray tube monitors for faces, which alight with a heart-shaped animated sprite as a show of affection. The cat presents these robots with a tangible connection to the emotional bonds that humans once formed with their pets.

Escaping ravenous sewer creatures

Morusque and the other denizens of the Slums are restricted from venturing outside their neighborhood enclave. Just outside the perimeter, a lair of carnivorous mega-bacteria known as Zurks awaits. Much of the action gameplay in *Stray*, situated outside of the inhabitable zones, revolves around outrunning these pint-sized, ravenous pests.

For the chase scenes, a different approach to sound design was required. Because the accompanying music cues were scored for tight, hair-raising setpieces, the musician and the designers had to deliver the expected thrills of a perilous encounter without falling back on cliché.

"I usually tend to think that it's not because you have monsters around that the music has to be stressful all the time," Van der Cruyssen says of the Zurk encounters. "I think music is allowed to speak about the environment. Usually when there were these chase sequences, I would start with a tune that is not necessarily fast-paced. But then I put it in the game and I usually got some requests from the team that it's not action-oriented enough. There were a lot of places in the game where I had to revise and put a lot more rhythm until they were happy with it."



Stray was built using Unreal Engine 4 without the use of middleware for audio, such as Wwise or FMOD. "Everything that was dynamic or required synchronizations, or any advanced audio logic, would require us to program an additional layer of logic for Unreal," the composer explains. "That was done for action sequences where several pieces of music were synchronized over a tempo, and that is the kind of thing that required a lot of debugging. The tunes would go out of sync, and sometimes we didn't understand why. There were problems like that."

As with the music cues paired with action sequences, some music tracks serve isolated puzzle segments. The synthetic sounding composition "Rikonium" features fixed arpeggios that evoke the feeling of reasoning through a problem.

By contrast, the way music operates during exploration segments is surprisingly dynamic. Most players might not recognize it until starting up a second playthrough, but the way in which the music score for *Stray* reveals itself, outside of the puzzle and action sequences, can be very unpredictable.

"There are two zones in the game that are a bit more open: the Slums and Midtown. For these zones, the way the music is handled is a bit strange," says Van der Cruyssen. "Two different players will not necessarily hear the same music at the same time. It's not random, but when the player does a specific action, if the action is important—the game may play a tune or not. And what tune it plays depends on what the action is, and whether there has been mu-

sic playing recently or not. Sometimes the game will choose to let it be silent."

Uncovering a hidden laboratory



During the course of development, BlueTwelve explored various possibilities for the robot inhabitants to connect with ancient cultural artifacts. Some of these ideas proved difficult to implement, such as a religious practices observed at an Asian temple that no longer appears in the game. While the developers drew inspiration from the Kowloon Walled City in Hong Kong, which represented an ideal urban environment for the cat to scale and explore, they ultimately decided against specifying the location of the robot city geographically.

One vestige that remains of the temple location is a music track called "Outlaws," influenced by Indonesian gamelan music. "Using typical gamelan scales with synths, instead of an actual gamelan, was one of the directions I took, at first, for some of the pieces," the composer says of the genre of "Outlaws", "but I was requested to make them sound less dissonant. In fact, the microtonality is mostly gone in the final versions, but I kept the overall mood of these pieces with an occasional wobbly pitch and it was enough to evoke that culture, in my opinion."

The lost human civilization of *Stray* resembles the crowded apartments of Kowloon, the disco halls of Europe, and so many other disparate trappings of far-flung urban settings. Similarly, the music score spans multiple genres, making it impossible for the listener to pin down one specific region of the

world. As a result, the city can best be described as 'unknown': post-human, existing at some unknowable point in the future.

Contrasted against the mechanistic structure of "Rikonium," "Raft" presents a strong, emotionally evocative melody. "This was initially one of the generic tunes that I put in the playlist for the Slums," says Van der Cruyssen, "and I observed people playing the Slums and getting this 'Raft' tune at various points. Sometimes they would get that when there was dialog, or when they discovered something. And the player didn't know that the song was not made on purpose for that particular moment."

"It's funny that sometimes they said, 'Oh, this tune is very good for this particular moment,' even though it was not supposed to happen there."

The capricious behavior of the soundtrack made playtesting a laboratory for new discoveries. The composer explains that he initially did not have plans for a pivotal scene that allows the player to progress beyond the Sewers to be attached to an emotional piece of music. "I thought I would make a dark, strange tune," he recalls.

"But having a soft tune playing actually made a lot of sense because there was this character who discovers there was this lab in his father's home... It was a choice that I made because it randomly appeared here during the playtest."

Infiltrating a midtown nightclub

In a Midtown setting beyond the dwelling of the Zurks, the cat is tasked with infiltrating a nightclub. For the plot to progress, B-12 must insert a dubstep tape into a cassette player at a nearby clothing store in an attempt to distract the shop owner. This sequence presented the composer with a difficult challenge: namely, finding a way to smoothly transition from the non-diegetic background track belonging to the score, to the diegetic track triggered by B-12 turning on the radio, and back again upon exiting the location.

"At some point, I knew that the player would have to enter a room where there would be a radio," the composer explains. "I adjusted the length of the music and set specific triggers to switch from the music to the outro of the music at the right time, so that the intradiegetic music would replace extra-

diegetic. But sometimes, for many reasons, the developer would decide to take the radio and put it somewhere else. I had to find another way of arranging the score around these elements."



Midtown scene from Maxim Dorokhov portfolio on [Artstation](#)

Since *Stray*'s release in mid-2022, dozens of players have uploaded music covers to YouTube, along with harder-to-find themes not included in the soundtrack album. "I've seen a few covers and tutorials on sheet music for the tunes and they are almost always slightly wrong," the composer observes, "but it's probably also due to the fact that the instruments are sometimes slightly out of tune or have some kind of resonance that makes them hard to actually pick out the notes.

"It had to be a bit organic, not in the biological sense, but in the way that there are a lot of glitches and imperfections... One thing that I tried to do, in general, in the game was to not have anything that reminds the player of humans too much. Even when there are actual instruments, like a violin, they are very processed so that you cannot always recognise the instrument."

One music track that exemplifies this approach to the score is the theme for the Guardian character, encountered in the Slums. The track was created using a drum machine and analog synths, recorded in several layers.

"If I recall correctly, it is one case where I put a lot of effects in parallel that would have an impact on the pitch so the instrument would be very dissonant — a lot more than what you hear in the final track — and then my job was just

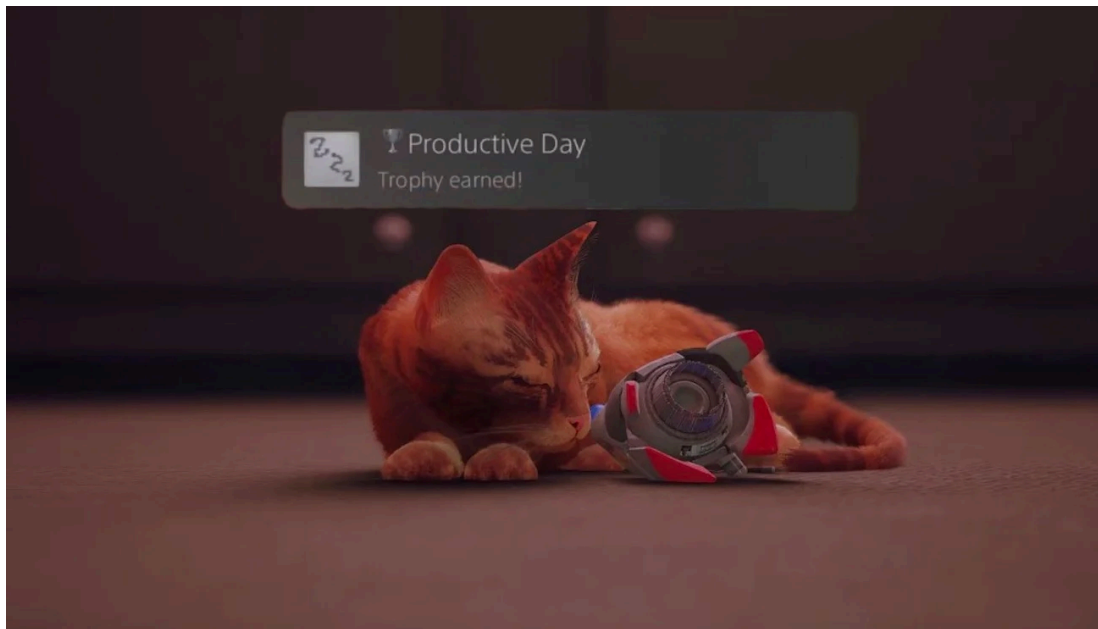
to cut the parts that were too dissonant so that it was okay to be heard. But I like this effect. You can hear that it's almost going to crumble, but never completely. Very unstable."

Pausing play for a catnap

As with the appearance of BlueTwelve's in-game avatars, the assortment of resting spots littered throughout the game was another last-minute addition. This optional feature treats the player to a respite between trotting across rooftops, wooden plank walkways, and city streets. Upon curling up for a catnap, the camera zooms out and a musical theme begins playing.

Upon first encountering this mechanic, the composer recalls, "I initially put some music in this spot as an easter egg. I had one piece of music that was not used in the game and I thought 'why not put it here?' I think it was a very hidden easter egg: you had to lie down for about one hour before the music started."

In the final build of the game, the player is rewarded for a one-hour nap not with a hidden piece of music but with a "Productive Day" Trophy. As the napping set-pieces were added, the composer was afforded new spots for unique compositions.



"I took a lot of tunes that were initially made for previous versions of the levels but were currently unused and I found a new place for most of them by making them triggered by these spots. I like that there is a lot of hidden mu-

sic triggered by random actions like this. If you go out of your way, you can explore a wide variety of music from the game."

Stray contains an enormous volume of audio for a game scored by a single composer. When it came time to publish the soundtrack, Van der Cruyssen's initial submission was over three hours in length.

The digital release on Steam narrowed the selection to just over two hours of extradiegetic score and intradiegetic street musician tunes and radio songs. The current record for speedrunning *Stray* clocks in at just under an hour; a feat that falls short of revealing the entirety of the music score.

"If you put every piece of music together, the full soundtrack is about five hours — it's not possible to hear it all in an hour, that's for sure!"



Yann van der Cruyssen is a composer and multidisciplinary creator –
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